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A Heterogeneous Hybrid CNN-Transformer Accelerator With Multi-Level Strassen MatMul

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ABSTRACT

A heterogeneous accelerator is proposed for efficient edge inference of hybrid CNN–Transformer vision networks. The architecture integrates a reconfigurable systolic array (SA), a vector processing element (VPE) array and a reduction core (RC) to support diverse computation patterns. The SA is reconfigured for standard, pointwise and depthwise convolution and one-level Strassen MatMul, while multi-level Strassen MatMul is jointly executed by the SA and VPE array to reduce the multiplication workload. Fixed-point nonlinear operators are processed by the VPE array and RC, enabling integer-only execution with high hardware utilization. Architectural modelling predicts up to 1.58× speedup over standard MatMul. FPGA measurements show 363.2 FPS on EfficientViT-B1 and 277.1 FPS on MobileViT-XS.

1 | Introduction

Vision transformers (ViTs) enable global image modeling through self-attention [1, 2], but their high computational and parameter costs limit edge deployment. Hybrid CNN–Transformer networks (HybridViTs) combine convolution for local feature extraction with Transformer blocks for global context modeling, providing an attractive solution for edge vision applications [3, 4].

Existing FPGA and ASIC accelerators mainly target standalone CNNs or Transformers [5–7]. However, HybridViTs combine convolution, self-attention and nonlinear operators with heterogeneous computation patterns, making efficient resource sharing difficult. EdgeCT [8] and FLASH [9] support hybrid CNN–Transformer networks, but their architectures provide limited resource sharing and reconfigurability. Moreover, critical operations such as large-scale MatMul and nonlinear computation remain challenging for efficient hardware implementation.

MatMul is a major computational bottleneck in edge inference. Strassen reduces eight multiplications to seven at the cost of additional additions and scheduling complexity [10]. Strassen-Hardware [11] uses seven parallel cores with pipelined addition and multiplication, but its recursive add/sub networks incur hardware overhead, while its MatMul-oriented datapath limits resource reuse across HybridViT operators.

In this letter, we propose a multi-core heterogeneous accelerator integrating a reconfigurable systolic array (SA), a vector processing element (VPE) array and a reduction core (RC) for HybridViT inference. The SA supports four modes and jointly executes multi-level Strassen MatMul with the VPE array, while the VPE array and RC handle nonlinear operators. For transposed MatMul, weight rearrangement during loading reduces additional cycles.

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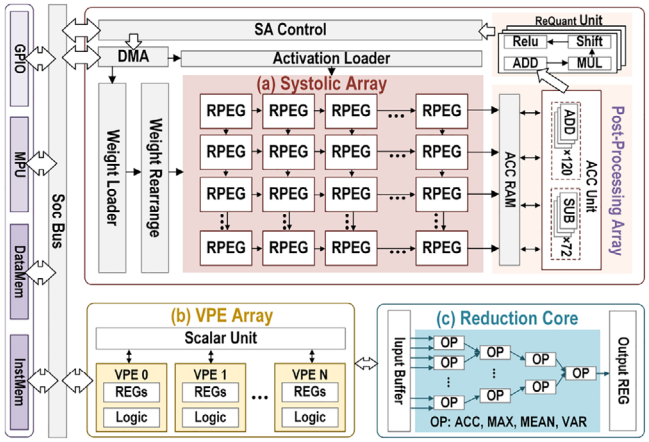


FIGURE 1 | Overall architecture of the proposed accelerator.

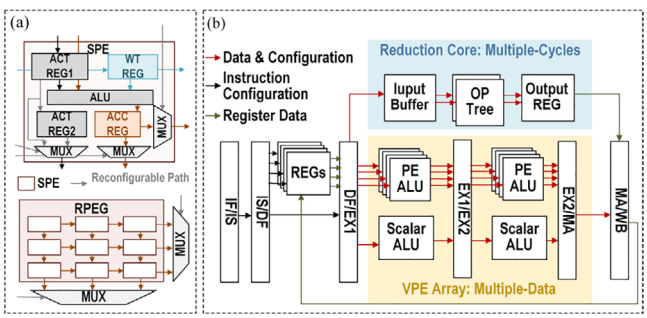


FIGURE 2 | Processing architecture and dataflow: (a) RPEG; (b) VPE array and RC.

2 | Multi-Core Hardware Architecture

HybridViT operators fall into three categories: (a) compute-intensive operations, including MatMul and convolution; (b) data-intensive operations involving repeated data accesses with simple computations, including post-processing and data rearrangement; and (c) cross-vector global operations, including vector-wise maximum and summation. Accordingly, MatMul and convolution are executed by the compute-intensive core with support from the data-intensive core, while nonlinear operators are jointly executed by the data-intensive and cross-vector global cores. The accelerator integrates SA, VPE array and RC, with dedicated control and memory modules, as shown in Figure 1.

The SA contains a controller, DMA, loaders, weight rearrangement, a 16×16 array of reconfigurable processing element groups (RPEGs), an accumulation (ACC) unit and a requantization unit. The controller manages dataflow and configuration, while DMA and loaders transfer data between DataMem and the RPEG array. The ACC unit buffers partial sums, the requantization unit performs output conversion and the weight rearrangement unit enables weight transposition during loading. As shown in Figure 2a, each RPEG contains a 3×3 array of systolic processing elements (SPEs), each with an 8-bit MAC, two activation registers, one weight register and one accumulation register. The weights remain stationary during computation, while activations propagate vertically and partial sums accumulate horizontally.

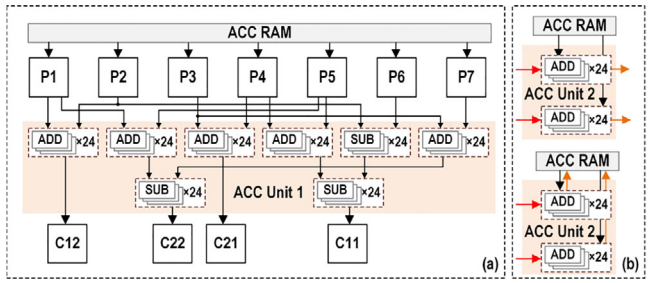


FIGURE 3 | ACC configurations for (a) Strassen MatMul and (b) convolution.

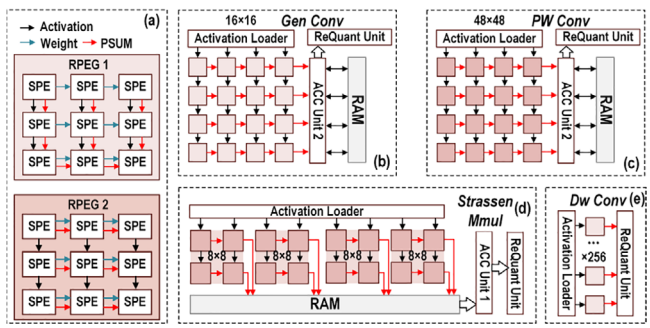


FIGURE 4 | Systolic-array configurations: (a) RPEGs; (b) standard convolution; (c) pointwise convolution; (d) Strassen MatMul; (e) depth-wise convolution.

The VPE array consists of a scalar unit and 256 vector processing elements (VPEs) supporting SIMD arithmetic and logic operations. Each VPE has an independent register group for parallel data processing. The reduction core (RC) comprises multiple operator trees, each reducing 32 input elements in parallel. In Figure 2b, the VPE array sends register data and control to the RC input buffer. The RC performs iterative reductions under single-instruction multiple-cycle (SIMC) control and returns results through output registers.

3 | Reconfigurable Computation of Convolution and MatMul

Convolution and MatMul require different dataflows and accumulation patterns. As shown in Equations (1) and (2), Strassen reduces eight multiplications to seven at the cost of extra additions and scheduling overhead. To accommodate these requirements, the systolic array supports two RPEG and ACC configurations, as shown in Figures 3 and 4a. RPEG-1 uses spatial partial-sum propagation for convolution, whereas RPEG-2 performs horizontal inner-product accumulation for pointwise convolution and MatMul. ACC Units 1 and 2 handle Strassen result recombination and cross-channel accumulation, respectively. Joint reconfiguration enables efficient resource reuse across operators without dedicated computing arrays.

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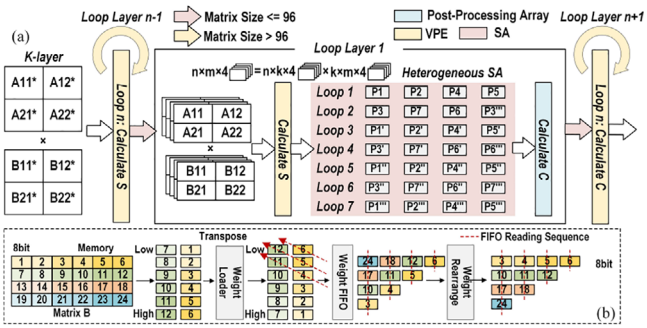


FIGURE 5 | (a) Multi-level Strassen MatMul; (b) weight loading for transposed MatMul.

$$\begin{cases} P_1 = A_{11}(B_{12} - B_{22}), \quad P_2 = (A_{11} + A_{12})B_{22}, \\ P_3 = (A_{21} + A_{22})B_{11}, \quad P_4 = A_{22}(B_{21} - B_{11}), \\ P_5 = (A_{11} + A_{22})(B_{11} + B_{22}), \\ P_6 = (A_{12} - A_{22})(B_{21} + B_{22}), \\ P_7 = (A_{11} - A_{21})(B_{11} + B_{12}). \end{cases} \tag{1}$$

$$\begin{cases} C_{11} = P_5 + P_4 - P_2 + P_6, \quad C_{12} = P_1 + P_2, \\ C_{21} = P_3 + P_4, \qquad\qquad C_{22} = P_5 + P_1 - P_3 - P_7. \end{cases} \tag{2}$$

For standard convolution, the systolic array is configured as a 16×16 array of RPEG-1 with ACC Unit 2, as shown in Figure 4b. Each RPEG performs spatial convolution using its 3×3 SPEs, while ACC Unit 2 accumulates partial sums across input-channel groups, enabling 16 input and 16 output channels to be processed in parallel. Pointwise convolution adopts RPEG-2 with ACC Unit 2, as shown in Figure 4c, where horizontal inner-product accumulation enables 48 input and 48 output channels to be processed per iteration. For depthwise convolution, the array is reconfigured into a 256×1 structure using RPEG-1, allowing 256 channels to be processed in parallel without cross-channel accumulation, as shown in Figure 4e.

For first-level Strassen MatMul, the systolic array is reconfigured into four 8×8 sub-arrays using RPEG-2 and ACC Unit 1, as shown in Figure 4d. The 1×4 sub-array configuration reduces the minimum supported matrix size, enabling deeper Strassen decomposition and higher speedup for smaller matrices. The seven products P_1–P_7 are computed according to Equation (1) and stored in the ACC RAM, after which ACC Unit 1 performs the required additions and subtractions to generate C_{11}–C_{22}. The computation is scheduled over seven iterations, with four matrix-multiplication blocks processed in parallel per iteration.

For larger matrices, multi-level Strassen is employed, as shown in Figure 5. The input matrices are recursively partitioned according to the selected recursion depth, and the required input sum/difference matrices are generated by the VPE array. The SA executes the base-level Strassen computations, while the VPE array performs the higher-level additions and subtractions to obtain the final C. This mapping assigns multiplication-intensive operations to the SA and addition/subtraction-intensive operations to the VPE array, reducing dedicated adder overhead and enabling concurrent execution. Accordingly, one-, two- and three-level Strassen are selected for $n \leq 96$, $96 < n \leq 192$ and $n > 192$, respectively. As shown in Figure 5b, fast matrix transposition is achieved during weight loading. Weight rearrangement and the FIFO directly provide the required weight-register layout, reducing additional transpose and rearrangement cycles. The proposed architecture is not limited to a 48×48 systolic array and can be scaled to different array sizes for different workloads.

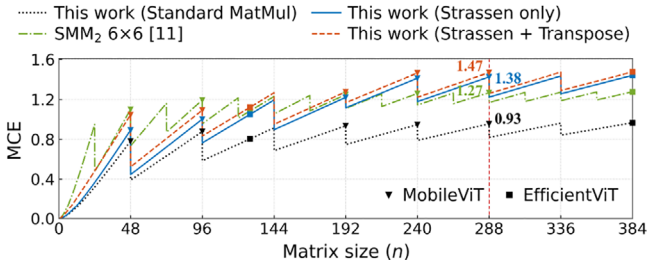


FIGURE 6 | Modelled MCE of the proposed architecture with different optimizations and SMM_2 6×6 [11]. Squares and inverted triangles mark matrix sizes of EfficientViT and MobileViT, respectively.

4 | Parallel Execution of Nonlinear Operations with VPE Array and RC

Fixed-point Softmax and LayerNorm are implemented through integer-only approximations and jointly executed by the VPE array and RC. For Softmax, the input is first shifted by its maximum value for numerical stability, and the exponential function is approximated using a linear function. The VPE array performs exponential approximation and normalization, while the RC computes the maximum and sum reductions. For LayerNorm, the RC computes the mean and variance, while the VPE array performs radix-2 square-root iteration, normalization and fixed-point scaling. These operations are decomposed into arithmetic and reduction operations, allowing the VPE array and RC to operate concurrently. An 8:1 array-size ratio balances execution cycles and improves hardware utilization.

5 | Implementation and Results

The multiplier compute efficiency (MCE) [11] is defined as $\text{mults}/(N_{\text{mul}}C)$, where mults denotes the conventional multiplication count, N_{mul} the multiplier count and C the modelled cycles including tiling, loading, Strassen additions/subtractions and transpose processing. Figure 6 shows that this work achieves higher MCE than [11] for larger matrices. With transpose optimization, the modelled speedup reaches 1.58× over standard MatMul at $n = 288$. Table 1 compares add/sub resources and ideal equivalent peak throughput. Both add/sub subsystems were synthesized for VCU118 with a 300-MHz target. Our design uses 192 ACC and 256 shared VPE add/sub paths; the reference network was reconstructed from [11]. At full utilization, ideal equivalent peak throughput is $2N_{\text{mul}}f(8/7)^L/10^9$ GOPS, with f in Hz and L the Strassen depth. Compared with [11], the proposed

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TABLE 1 | Add/sub resources and ideal peak throughput.

Design	Multipliers	Add/sub per mult.	LUT	FF	CARRY8	Ideal eq. peak (GOPS)
SMM_2 6×6 [11]	1764	0.551	32,089	26,872	4896	1382.4
This work	2304	0.194	22,276	15,884	1280	2063.5

TABLE 2 | Quantized model accuracy and inference FPS.

	Accuracy (%)			FPS		
Network	FP32	INT16 NL	INT8	No opt.	Transpose	Transpose+Strassen
MobileViT-XS	73.31	73.17	72.91	256.4	263.7	277.1
EfficientViT-B1	79.05	78.96	78.04	344.0	350.7	363.2

TABLE 3 | Comparison with related and comparable works.

	TCAS-I'23[6]	TVLSI'26[7]	ICFPT'25[8]	TNNLS'26[9]	This work	
Technology	16 nm FPGA	16 nm FPGA	16 nm FPGA	16 nm FPGA	16 nm FPGA	
Model Type	Trans.	Trans.	CNN+Trans.	CNN+Trans.	CNN+Trans.	
Frequency	300 M	300 M	200 M	100 M	300 M	
BRAM	—	383.0	1704.0	1141.0	404.0	
LUT	144.0 K	—	194.7 K	350.9 K	256.1 K	
DSP	1268	1754	676	1180	778	
Model	ViT-Tiny	ViT-Tiny	CTFormer	MobileViT-XXS	MobileViT-XS	EfficientViT-B1
GOPS	616.1	455.3	457.7	—	609.6	508.5
FPS	245.3	182.1	—	99.8	277.1	363.2
Power (W)	—	9.92	—	6.31	10.642	11.066
GOPS/DSP	0.486	0.260	0.677	—	0.784	0.654
GOPS/W	—	45.9	—	—	57.3	46.0

design reduces add/sub hardware resources while achieving higher ideal peak throughput.

MobileViT-XS and EfficientViT-B1 provide complementary workloads covering convolution, Softmax attention, feature unfolding/folding and multi-scale ReLU linear attention [3, 4]. They also cover the main computation types in ViT-Tiny and CTFormer [6–8], while differing in operator mix and matrix shapes. Therefore, GOPS is used for cross-model throughput comparison, with FPS reported for each workload. FPS and GOPS are derived from on-board timing on VCU118 at 300 MHz. Power is estimated using Vivado post-route report power, and GOPS/W is calculated from measured throughput and estimated power.

We evaluate quantized model accuracy on ImageNet-1K [12] and inference FPS across optimizations. Table 2 shows that 16-bit fixed-point nonlinear operators cause Top-1 accuracy losses of only 0.14 and 0.09 percentage points, respectively, relative to FP32. Further quantizing convolution and MatMul to INT8 increases total accuracy losses to 0.40 and 1.01 percentage points. Meanwhile, transpose optimization improves FPS by 2.85% and 1.95%, while Strassen provides gains of 5.08% and 3.56% over the transpose-only configuration. These results demonstrate improved end-to-end throughput with limited accuracy loss.

Table 3 compares the proposed accelerator with related designs. Unlike accelerators dedicated to CNNs or Transformers, the proposed architecture supports both workloads through shared reconfigurable hardware. Compared with EdgeCT [8] and FLASH [9], it further integrates convolution, MatMul and nonlinear operators, while multi-level Strassen reduces the multiplication workload. The accelerator achieves 609.6 GOPS and 277.1 FPS on MobileViT-XS, and 508.5 GOPS and 363.2 FPS on EfficientViT-B1. These results demonstrate efficient execution across different hybrid CNN–Transformer workloads.

6 | Conclusion

This letter presents a multi-core heterogeneous accelerator for HybridViTs, integrating a reconfigurable systolic array, VPE array and reduction core to support convolution, MatMul, and nonlinear operators. Multi-level Strassen MatMul is jointly implemented by the systolic and VPE arrays, while reconfigurable

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dataflows enable efficient hardware reuse across CNN and Transformer workloads. The architectural model gives a 1.58× MatMul speedup at $n = 288$, while FPGA evaluation demonstrates real-time inference on multiple HybridViTs.

Author Contributions

Yuxiao Wen: investigation, methodology, writing – original draft. **Wenyao Zhao:** software, validation. **Xu Yang:** formal analysis. **Zhe Wang:** funding acquisition. **Zhao Zhang:** data curation. **Jian Liu:** resources. **Nanjian Wu:** supervision. **Runjiang Dou:** conceptualization, writing – review and editing. **Liyuan Liu:** project administration, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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